higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MECHANOTECHNICS N4

1 APRIL 2019

This marking guideline consists of 7 pages.

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QUESTION 1

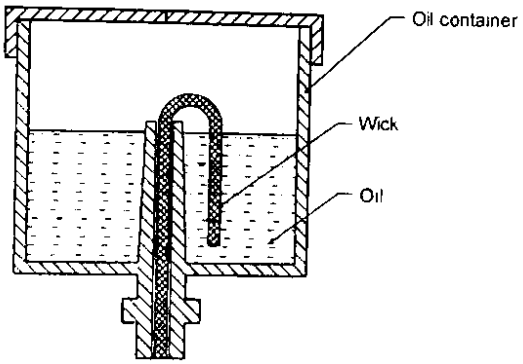
1.1 With individual production you concentrate all manufacturing process efforts on one item.✓✓ Workers complete all operations on this single item before they start working on the next.✓ (3)

1.2
- Manufacturing a boiler in a workshop where one item can be made at a time
- When a workshop is instructed by a client to cast or machine a special casting (2 x 1) (2)

1.3
- Only one product is produced at a time.
- Specialised equipment is needed.
- Layout is flexible.
- Capital outlay is extremely high.
- The workers are flexible and highly skilled. (5 x 1) (5)

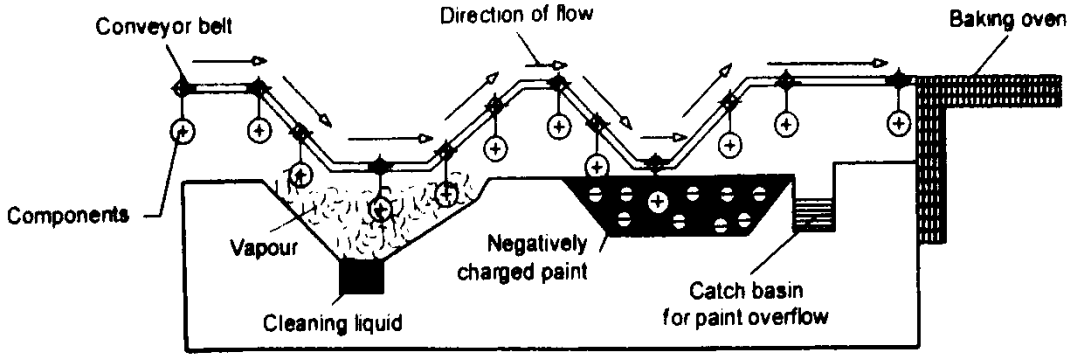
1.4
- Excessive lubrication
- Mounting the bearing the wrong way
- Mounting the bearing in a damaged housing or shaft
- Applying force to the wrong ring during mounting
- Misaligning the bearing during mounting
- Mounting the housing and cover incorrectly (Any 5 x 1) (5)

1.5



(5)

1.6



(5)

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1.7
- Excessive use of paint
- Large quantities of paint are kept in paint dip troughs
- Loss of thinners through evaporation
- Only for mass production
- If paint is left in dip troughs overnight, continual, thorough agitation is needed the following day. (5 x 1) (5)

[30]

QUESTION 2

$$v=\frac{\pi . D . N}{60}$$
$$=\frac{\pi x\, 0,25\, x\, 1800}{60} \checkmark$$
$$=23,562\, m/s \checkmark$$
$$T_C = mv^2$$
$$=0{,}5(23,562)^2 \checkmark$$
$$=277,584\, N \checkmark$$
$$\frac{T_1 - T_C}{T_2 - T_C} = e^{\mu\theta}$$
$$\frac{900-277,584}{T_2-277,584} = e^{0,3\, x\, 2,792} \checkmark$$
$$\frac{622,416}{T_2-277,584} = 2,311 \checkmark$$
$$622,416 = T_2\, 2,311 - 641,497 \checkmark$$
$$T_2 = \frac{1263,913}{2,311} \checkmark$$
$$\therefore T_2 = 546,912\, N \checkmark$$
$$(T_1 - T_2)n = \frac{P}{v}$$
$$(900-546,912)n = \frac{30\, x\, 10^3}{23,562} \checkmark\checkmark$$
$$n = \frac{30000}{8319,459} \checkmark$$
$$=3,606 \checkmark$$
$$say\ n = 4\ belts \checkmark$$

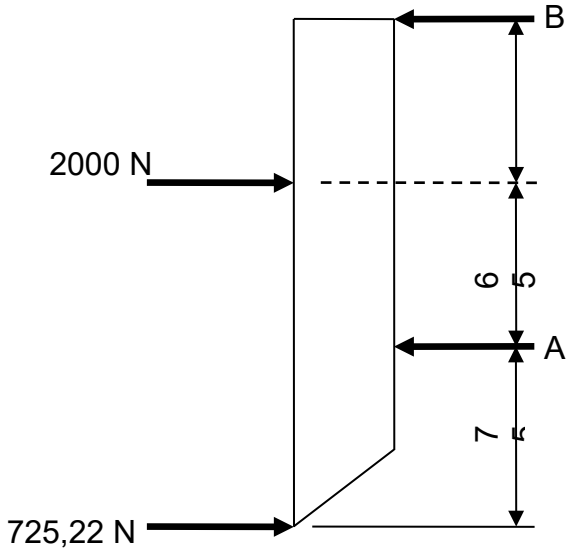
[14]

QUESTION 3

3.1
$$F_f = \mu x m x g$$
$$F_f = 0{,}3 \; x \; 150 \; x \; 9{,}81$$
$$= 441{,}45 \; N \; \checkmark$$
$$WD_f = F_f \; x \; S_l$$
$$= 441{,}45 \; x \; 0{,}15 \; \checkmark$$
$$= 66{,}218 \; J \; \checkmark$$
(3)

3.2
$$WD_{total} = WD_{cutting} + WD_{friction}$$
$$WD_{cutting} = WD_{total} - WD_{friction}$$
$$= 175 - 66{,}218$$
$$= 108{,}783 \; J \; \checkmark$$
$$F_{cutting} = \frac{WD_{cutting}}{S_l}$$
$$F_{cutting} = \frac{108{,}783}{0{,}15} \; \checkmark$$
$$= 725{,}22 \; N \; \checkmark$$
(3)

3.3



Take moments about B to find A
$$\sum CWM = \sum ACWM$$
$$Ax130 = (725{,}22 \; x \; 205) + (2000 \; x \; 65) \; \checkmark\checkmark$$
$$A = \frac{278670{,}1}{130} \; \checkmark$$
$$= 2143{,}616 \; N \; \checkmark$$

Take moments about A to find B
$$\sum CWM = \sum ACWM$$
$$2000 \; x \; 65 = (725{,}22 \; x \; 75) + (Bx130) \; \checkmark\checkmark$$
(8)

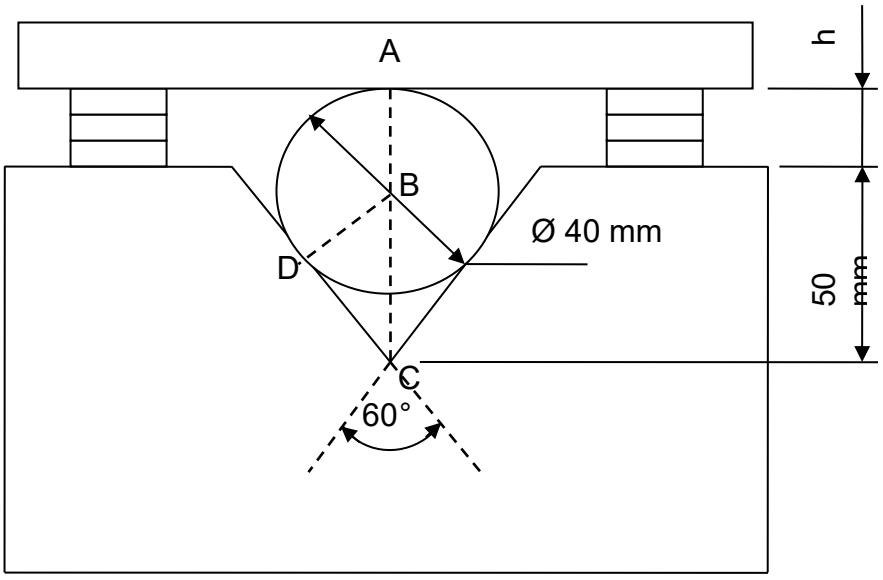
$$B = \frac{75608,5}{130} \checkmark$$
$$= 581,604\,N \checkmark$$

[14]

QUESTION 4

4.1 1 – Straight edge
2 – Gauge block
3 – Roller
4 – V-block (4)

4.2



$$h = AC - 50$$
$$AC = AB + BC$$
$$BC\ in\ BCD$$
$$Sin\,\theta = \frac{BD}{BC}$$
$$BC = \frac{20}{\sin 30°} \checkmark$$
$$= 40\,mm \checkmark$$
$$AC = AB + BC$$
$$AC = 40 + 20 \checkmark$$
$$N_D = 1000\,r/\min \checkmark$$
$$h = AC - 50$$
$$= 60 - 50 \checkmark$$
$$= 10,\,mm \checkmark$$

(6)

4.3
- Place the 40 mm diameter precision roller gauge.
- Set up two sets of gauge blocks exactly 10 mm thick.
- Place a set of gauge blocks on either side of the groove and place the straight edge on top.
- The angle of the groove will be correct if the two sets of gauge blocks (4)

each have a thickness of 10 mm and the straight edge touch the top of the roller.

[14]

QUESTION 5

5.1

Condition	Gear A	Gear B	Gear C	Gear D	Arm E
Fix arm E and rotate Gear A by +1rev	$+1\surd$	$\frac{T_A}{T_B}=\frac{100}{16}$ $=6,25\surd$	$=6,25$	$\frac{T_A}{T_B}x\frac{T_C}{T_D}$ $\frac{100}{16}x\frac{22}{-28}$ $=-4,911\surd$	$0\surd$
Multiply by x and add y	$x+y\surd$	$6,25x+y$ $\surd$	$6,25x+y$	$-4,911x+y\surd$	$+y\surd$
$N_E=0$ $N_D=1000r/\min$ $N_A=?$	$N_A=?$ $\surd$			$N_D=1000r/\min$ $\surd$	$N_E=0$ $\surd$
$N_A=0$ $N_D=1000r/\min$ $N_E=?$	$N_A=0$ $\surd$			$N_D=1000r/\min$ $\surd$	$N_E=?$ $\surd$

(7)

$x+y=0.....1\checkmark$

$y=0.....2$

substitute …2 in ….1

$-4,911x+y=1000.....1$

$x=\frac{1000}{-4,911}$

$x=-203,625\checkmark$

$x+y=N_A$

$N_A=-203,625+0$

$N_A=-203,625r/\min\checkmark$

(3)

5.2

$x+y=0$

$x=-y.....1$

$-4,911x+y=1000.....2\checkmark$

substitute …1 in ….2

$-4,911(-y)+y=1000\checkmark$

$y=\frac{1000}{5,911}$

$y=169,176\checkmark$

(4)

$y = N_E$

$N_E = +169,176\,r/\min$ ✓

[14]

QUESTION 6

$$Q_1 = v_1 x A_1$$

$$v_1 = \frac{Q_1}{A_1}$$

$$v_1 = \frac{0,09}{\pi/4(0,1)^2} \checkmark\checkmark$$

$$v_1 = 11,46\, m/s \checkmark$$

$$Q_2 = v_2 x A_2$$

$$v_2 = \frac{Q_2}{A_2}$$

$$v_2 = \frac{0,09}{\pi/4(0,18)^2} \checkmark\checkmark$$

$$v_2 = 3,6\, m/s \checkmark$$

$$\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + h_1 = \frac{P_2}{\rho g} + \frac{v_2^2}{2g} + h_2$$

$$\frac{230000}{10^3 x 9,81} + \frac{(11,46)^2}{2x9,81} + 0 = \frac{P_2}{10^3 x 9,81} + \frac{(3,6)^2}{2x9,81} + 15 \checkmark\checkmark$$

$$23,445 + 6,693 = \frac{P_2}{9810} + 0,66 + 15 \checkmark\checkmark$$

$$\frac{P_2}{9810} = 30,138 - 15,66 \checkmark\checkmark$$

$$\frac{P_2}{9810} = 14,478 \checkmark$$

$$P_2 = 142029,18\, Pa$$

$$P_2 = 142,029\, kPa \checkmark$$

[14]

TOTAL: 100

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